

Ajitkumar Senthil Kumar

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[Portfolio](#) | [LinkedIn](#) | [Github](#)

Summary

Full-stack & AI/ML engineer (2+ years) building production web apps and RESTful APIs in Python (FastAPI) and React/TypeScript. Ship scalable features with testing, CI/CD (GitHub Actions), Docker, SQL, and Cloud services. Automate data pipelines and integrate ML inference to elevate system performance and UX.

Education

State University of New York at Buffalo | *M.S., Data Science (GPA: 3.3/4)* 2024 - 2025

Anna University, Chennai | *B.E., Mechanical Engineering (GPA: 8.1/10)* 2017 - 2021

Experience

Human CloudSoft | *Software Intern* May 2023 - Jun 2024

- Developed a web-based inventory platform that automated stock tracking, transitioning all stock logs from manual to digital.
- Enhanced inventory accuracy by 35% and reduced audit time by 40% by implementing automated alerts and streamlining workflows.

Klicknet Info Services Pvt Ltd | *Frontend Developer* Apr 2022–Apr 2023

- Engineered responsive and accessible UIs using React and TypeScript, increasing user retention by 25%.
- Collaborated with backend teams to optimize workflows and performance, improving cross-team efficiency by 30%.

L&T Rubber Processing Machinery | *Design Engineer* Nov 2021–Feb 2022

- Converted complex 3D models into precise 2D AutoCAD drawings to ensure production readiness and defect-free outputs.
- Reduced manufacturing errors by 15% and rework by 20% through detailed component modeling in CREO and application of GD&T principles.

Projects

SmartLoanPred | *Full-stack ML & AI Application* Sep 2025–Present

- Built and deployed a **Full-stack, real-time loan-approval (classification and Regression) system with LLM report** (React + Tailwind, FastAPI) with JWT/OAuth2 with Role Context; Dockerized with CI/CD (GitHub Actions) and NGINX reverse proxy.
- Developing a **RAG-based Loan Assistant** for loan-specific Q&A (retrieval + model inference).

MIDAS | *Full-stack Deep Learning ,AI Application for HealthCare* Jul 2025-Present

- Developed an end-to-end pipeline in PyTorch for dermatology image classification and risk screening, applying extensive image preprocessing, and class-weighted loss to improve performance on a diverse dataset.
- Architected the transition from a research model to an ongoing product by integrating **Grad-CAM** as a core feature to ensure interpretability and trust for medical professionals, with **Agentic-RAG model for Doctor's scoped Medical Response**.

CovDet | *Full-stack Deep Learning ,AI Application for HealthCare* May 2025- Sep 2025

- Built a computer vision classifier to detect respiratory diseases from X-ray images, performing extensive image cleaning and augmentation while utilizing early stopping and cosine annealing to train robust, high-performance models.
- Engineered a production-ready inference pipeline by exporting final models to efficient TorchScript/ONNX formats and deploying them via a FastAPI endpoint for real-time analysis.

New York Taxi Rides | *MLOps* Feb 2024–Apr 2024

- Modeled NYC taxi ride demand using historical trip data through extensive feature engineering with lag variables, rolling windows, and external joins (holidays, weather).
- Used time-based cross-validation to ensure model robustness and integrated results into a Hopworks feature store.
- Developed a Streamlit dashboard for real-time monitoring and automated retraining workflows using GitHub Actions.

Technical Skills

- Programming:** Python, R, SQL, JavaScript, TypeScript
- ML/AI:** Scikit-learn, PyTorch, Pandas, NumPy, imbalanced-learn, MLflow, Jupyter
- Frontend:** React, Next.js, Tailwind CSS, shadcn/ui
- Backend:** FastAPI, SQLAlchemy, Firebase(OAuth)
- Databases:** PostgreSQL, SQL Server, Supabase, SQLite
- DevOps/Cloud:** Docker, GitHub Actions, AWS (Sagemaker,S3,Lambda), DigitalOcean
- RAG/Agents:** LangChain, FAISS/Chroma, Ollama